

< Previous











Next >

2. Matrix exponential

 Bookmark this page

Notifications

×

- Upgrade your course today**
- ✓ You are taking the free version of this course.
 - ✓ Upgrade today and, upon passing, receive your digital certificate signed by MIT faculty to highlight the knowledge and skills you’ve gained from this MITx course.

Upgrade

LTI Consumer (External resource) (1.0 points possible)

Using matrix exponential (expm)

Recall that for a linear system

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$$

with initial conditions

$$\mathbf{x}(0) = \mathbf{x}_0$$

we can write the solution in terms of the matrix exponential

$$\mathbf{x}(t) = e^{\mathbf{A}t}\mathbf{x}_0$$

This is often impossible to write down by hand, but Matlab has its function which can calculate matrix exponential. For a matrix $\mathbf{A}$, we can define $\mathbf{B} = e^{\mathbf{A}}$ in MATLAB by

```
B = expm(A);
```

Let's return to the example of three tanks connected tanks, where tank 1 has a leak, from recitation some volume of one fluid, and tank 3 contains the same volume of another fluid. These two tanks are connected by pipes of equal width. Tank 2 is initially empty. You found that all of the eigenvalues are negative. This means the system is stable.

In the next problem, you will show that you can create any mixture in the second tank, but you stop when you make sure you maximize the yield.

- 1. Use the template below to plot the solution for the system

$$\dot{\mathbf{x}} = \begin{bmatrix} -(a + \mu) & a & 0 \\ a & -2a & a \\ 0 & a & -a \end{bmatrix} \mathbf{x}$$

with $a = 1$, $\mu = 0.5$ and the initial conditions:

$$\mathbf{x}_0 = \begin{bmatrix} 2/3 \\ 0 \\ 1/3 \end{bmatrix}$$

- You should find the analytic solution by calculating matrix exponentials.
- 2. Estimate the time (to 2 significant figures) at which we should stop the apparatus running so that the concentration in tank 2 is 0.5. You should search in the range $t \in [0.5, 1.5]$.

Script ?

Save

Reset

MATLAB Documentation (https://www.mathworks.com/help/)

```
1 %We've set the values of a and mu as
2 %specified in the problem for you.
3 a = 1.0;
4 mu = 0.5;
5 %Enter the matrix A as defined above,
6 % the initial value vector x0, and create
7 % a row vector t of 400 equally spaced
8 %time values as t ranges from 0.5 to 1.5;
9 A = [-1.5 1 0; 1 -2 1; 0 1 -1];
10 x0 = [2/3;0;1/3];
11 t = linspace(0.5, 1.5, 400);
12 %At each time t, define x in terms of the exponential matrix expm(A), t, and x0
13 for m=1:length(t)
14     x(:,m) = expm(A*t(m))*x0;
15 end
16 %Let h1 be the height of fluid in tank 1.
17 %Let h2 be the height of fluid in tank 2.
18 %Let h3 be the height of fluid in tank 3.
19 %Define h1, h2, and h3 in terms of x.
20 h1 = x(1,:);
```

21 h2 = x(2,:);
22 h3 = x(3,:);
23 %Plot h1, h2 and h3
24 plot(t,h1,t,h2,t,h3)
25 %Determine the time tmax when the height in tank 2 is a maximum.
26 %Hint: look up the function max() in the matlab documentation.
27
28 [m, tm] = max(h2);
29 tmax = t(tm);
30

▶ Run Script

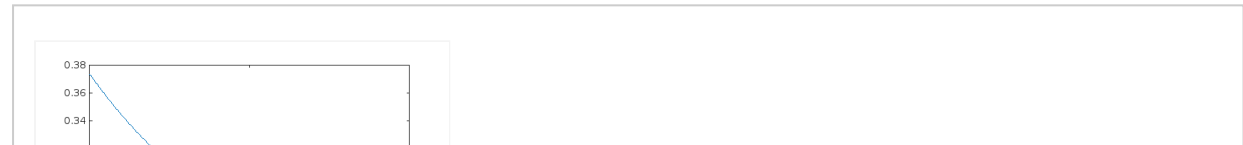
?

Assessment: All Tests Passed

Submit ?

- ✔ Check if A is correctly defined
- ✔ Check if x0 is correctly defined
- ✔ Check if t is correctly defined
- ✔ Check if h1 is correct
- ✔ Check if h2 is correct
- ✔ Check if h3 is correct
- ✔ Time when h2 is maximized found

Output



2. Matrix exponential

Hide Discussion

Topic: Unit 3: Solving systems of first order ODEs using matrix methods / 2. Matrix exponential

Add a Post

Show all posts

▼

by recent activity

▼

There are no posts in this topic yet.

✕

◀ Previous

Next ▶